

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2013-14

Paper: GLB-401 — Palaeontology and Stratigraphy

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem IV/

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Define Palaeontology and describe its relationship with other branches of geology. [14]

Question 2. Briefly discuss **any two** of the following: [2 × 7 = 14]

- (a) Essential conditions of fossilization
- (b) Origin of life
- (c) Various terms used to describe the bathymetric distribution of organisms

Question 3. Describe the morphological features of Trilobites. Support your answer with neat, neatly labeled diagrams. [14]

Question 4. Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) Incompleteness of the fossil record
- (b) Geological Time Scale
- (c) Economic importance of the Cuddapah Group

SECTION — B

Question 5. Describe in detail the various principles of stratigraphy. Illustrate your answer with neat diagrams. [14]

Question 6. Describe in detail the distribution, classification, and economic importance of the Dharwar Supergroup. Briefly discuss what kind of geological information is derived from these rocks. [14]

Question 7. Write short notes on **any two** of the following: [2 × 7 = 14]

- (a) In what way stratigraphy requires inputs from all branches of geology
- (b) What is the significance of the Eparchaeon Unconformity

- (c) Why unconformities are often marked by conglomerates

SECTION — C

Question 8. Distinguish between or answer in 1 to 3 sentences:

[7 × 2 = 14]

- (i) Fossils and pseudofossils
- (ii) Species and subspecies
- (iii) Sargur Group
- (iv) *Chondrites* and dendrites
- (v) Diamondiferous conglomerate
- (vi) Lithostratigraphic units
- (vii) Apical disc in Echinoids